

The empirical recurrence for OEIS A239173: recovering a conjecture that is not written down

Adrian Perez Fontelles
Independent researcher

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Abstract

OEIS A239173 records “Empirical recurrence of order 67”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 118$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 67, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 67 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A239173 is “Number of $(n+1) \times (3+1) 0..2$ arrays with no element greater than all horizontal neighbors or equal to all vertical neighbors.”. It has offset 1 and begins

98, 514, 9604, 130890, 1876940, 26726740, 381984614, 5442841504, ...

The entry states:

Empirical recurrence of order 67 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 10:58:34, revision 7) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 506$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 118$ of the 506, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 118$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 1012$ exact terms of the model returns order 67 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{67} c_i a(n-i),$$

c_1	4	c_{18}	54828099937	c_{35}	933491040210	c_{52}	70153167576
c_2	57	c_{19}	−147760515101	c_{36}	−961188644526	c_{53}	−4333233452
c_3	1316	c_{20}	269034944534	c_{37}	994295619698	c_{54}	12378062371
c_4	875	c_{21}	−557271530102	c_{38}	−2920809019442	c_{55}	12652986089
c_5	−4007	c_{22}	1091365174862	c_{39}	−1387998979757	c_{56}	−1138211851
c_6	−239679	c_{23}	−1350599240692	c_{40}	952157892420	c_{57}	2025857147
c_7	−62517	c_{24}	2035499302106	c_{41}	−1736197036599	c_{58}	−126204369
c_8	195215	c_{25}	−4144659067917	c_{42}	2692090998195	c_{59}	−412966715
c_9	14848671	c_{26}	5543099157312	c_{43}	−1526087465076	c_{60}	−44301927
c_{10}	2416115	c_{27}	−4696542675145	c_{44}	839243037053	c_{61}	−73353409
c_{11}	−30262689	c_{28}	8398348457072	c_{45}	747298996976	c_{62}	−2858349
c_{12}	−322221611	c_{29}	−12203829452504	c_{46}	−203778404598	c_{63}	5144543
c_{13}	−240207027	c_{30}	6942227866367	c_{47}	166890686647	c_{64}	−1114648
c_{14}	1598446602	c_{31}	−8788585601681	c_{48}	−470510430512	c_{65}	−1462666
c_{15}	485073553	c_{32}	10245876048569	c_{49}	13905511093	c_{66}	−366352
c_{16}	9384831444	c_{33}	−3909813985123	c_{50}	−53364100624	c_{67}	−19530
c_{17}	−30963545151	c_{34}	3185352182867	c_{51}	−84899388155		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 67, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $67 + 4$ terms removed returns order 67 as well, so $\deg q_{\min} = 67 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 16 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $67 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 67 that this sequence satisfies — and that family turns out to have exactly one member.

References

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